

What Are Market Regimes? Definitional Chaos, Validation Failure, and a Path Toward Methodological Convergence

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Abstract—Market regimes, persistent states of market behaviour characterised by distinct statistical properties, have attracted sustained research attention, yet the field has failed to converge on shared definitions, validated methodologies, or agreed empirical standards. This paper critically surveys 98 peer-reviewed papers published between 1989 and 2025, taxonomising them across five paradigm families: Markov/Hidden Markov Model approaches (43.9%), Hybrid methods (24.5%), Clustering (17.3%), Threshold/Rule-Based methods (7.1%), and Deep Representation Learning (7.1%). Three structural failures are documented. First, definitional chaos: only 1 of 98 papers provides an explicit, operational regime definition, with the remainder embedding distinct implicit conceptualisations spanning trend, volatility, macroeconomic, liquidity, and structural-break frameworks that are mutually incommensurable across temporal scales and causal mechanisms. Second, a validation crisis: no paper in the corpus simultaneously satisfies out-of-sample testing, robustness checks, look-ahead bias avoidance, transaction cost modelling, and code or data availability; 35 papers (35.7%) report no out-of-sample evaluation, and 83 papers (84.7%) ignore transaction costs entirely. Third, a reproducibility deficit: only 16 papers (16.3%) release code or data. The paper advances the thesis that regime detection has proliferated without converging because it systematically conflates ontological, epistemological, and pragmatic questions that remain unresolved, and proposes tiered reproducibility standards, asset-class-specific evaluation benchmarks, and a mandatory validation funnel as prerequisites for scientific maturation.

Index Terms—Market regimes, regime switching, Hidden Markov Models, financial time series, market volatility, systematic review, methodology comparison, empirical validation.

I. INTRODUCTION: WHY MARKET REGIME DETECTION HAS NOT CONVERGED

Market regimes, understood broadly as persistent states of market behaviour characterised by distinct statistical properties, have captivated financial researchers for decades. The intuition is compelling: if regime shifts could be reliably detected in real-time, they would enable adaptive portfolio allocation [1], [2], tactical trading strategies [3], [4], and dynamic risk management [5], [6]. Yet despite hundreds of papers and increasingly sophisticated models, regime detection remains stubbornly resistant to scientific consensus. This survey confronts an uncomfortable reality: after decades of research, we lack agreement on what regimes fundamentally are, how they should be identified, whether different methods measure the same underlying phenomenon, or even which

approaches work better than others. The field has proliferated without converging, raising a critical question: why has regime detection failed to mature into a reliable, generalisable science?

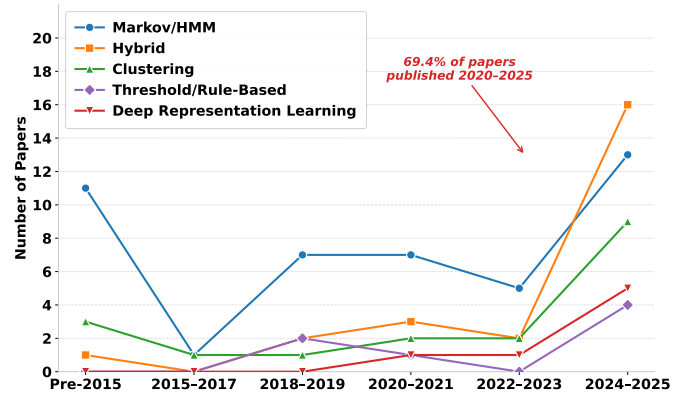


Fig. 1: Publication growth by paradigm family (1989 to 2025). Lines trace paper counts across six time periods for each paradigm family. The 2024 to 2025 period accounts for 47 of 98 papers (48.0%), with Hybrid and Deep Learning methods growing fastest, evidence of proliferation without convergence.

A. The Proliferation of Regime Models

Our systematic review of 98 papers reveals explosive acceleration (Figure 1): 68 papers (69.4%) published from 2020–2025 versus 14 papers in 2015–2019 and 14 papers before 2015, with 2 papers undated. Papers were identified through database searches of Google Scholar, SSRN, and arXiv using terms including “market regime detection,” “regime switching financial markets,” “Hidden Markov Model finance,” and related variants. Search results were supplemented by backward citation tracing from key survey papers. The observed growth from 14 papers (2015–2019) to 68 papers (2020–2025) likely reflects genuine field expansion, though we acknowledge this could be partially influenced by recency bias in search strategies. We exclusively review academic publications; practitioner methods are noted where evidence from industry surveys is available, but are not systematically included in the corpus. Yet this volume surge has not brought methodological convergence. The landscape fragments across five major paradigm families: Markov/HMM approaches (43 papers, 43.9%) [7]–[17], Hybrid methods (24 papers, 24.5%)

List of Abbreviations

Abbreviation	Full Form
AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion
bps	Basis Points
CBOE	Chicago Board Options Exchange
ELBO	Evidence Lower Bound
EM	Expectation–Maximisation
FX	Foreign Exchange
GARCH	Generalised Autoregressive Conditional Heteroskedasticity
HMM	Hidden Markov Model
i.i.d.	Independently and Identically Distributed
KL	Kullback–Leibler Divergence
LSTM	Long Short-Term Memory
MSM	Markov-Switching Model
VAE	Variational Autoencoder
VIX	CBOE Volatility Index

[18]–[26], a meta-category whose papers combine two or more of the other four paradigms and are counted here exclusively rather than in their constituent families, Clustering (17 papers, 17.3%) [27]–[33], Threshold/Rule-Based methods (7 papers, 7.1%) [34]–[38], and Deep Representation Learning (7 papers, 7.1%) [39]–[43]. These are exclusive counts: each paper is counted in exactly one category. Market coverage spans equities (76 papers) [2], [15], [21], [44]–[46], cryptocurrencies (15 papers) [47]–[51], foreign exchange (25 papers) [13], [34], [52]–[55], commodities (16 papers) [56], and multi-asset portfolios (10 papers) [57]–[66].

Most strikingly, our analysis identifies at least 98 distinct conceptualisations of what constitutes a regime across the surveyed papers. Papers define regimes as trend direction, latent statistical states with distinct mean returns and volatility [7], [8], volatility levels and market stress [14], [39], macroeconomic states [4], [9], [67], or microstructural order flow dynamics [68]. It is worth noting that papers sharing surface-level definitions may still differ substantially in operationalisation, estimation methodology, or empirical scope, meaning that counting unique conceptualisations likely provides an upper bound on true heterogeneity. Nonetheless, the degree of fragmentation is substantial: papers rarely share explicit definitions, draw on different theoretical foundations, and do not validate against common benchmarks.

B. Historical Roots: A Field Reinventing Itself

The current fragmentation is not novel; it echoes unresolved debates spanning more than half a century. Kondratiev waves (1920s), Burns and Mitchell’s NBER business cycle dating (1946), and chartist “trend identification” are all forms of regime detection under different names. The core tension between objective algorithms and expert judgement that fractured the NBER dating programme in the 1970s is methodologically identical to today’s HMM-versus-clustering debates.

Hamilton’s [69] Markov-switching model for GDP (1989) established the modern statistical framework, yet it took over a decade for the approach to migrate systematically to financial markets [7], [15]. The intervening period saw an important early critique. Diebold and Rudebusch [70] characterised much early regime work as “measurement without theory,”

arguing that statistical identification of switching states did not constitute economic explanation. This critique remains unaddressed in the contemporary literature. An authoritative survey by [2] confirmed that the proliferation of Markov-switching applications had not resolved foundational questions about what regimes represent. Today’s papers detect regimes with increasing sophistication but decreasing theoretical motivation, perpetuating exactly the pattern Diebold and Rudebusch identified.

C. Three Unresolved Questions

This survey advances a critical thesis: regime detection has failed to converge because the field systematically confuses three distinct questions that remain unresolved.

The Ontological Question: What are market regimes? Are they objective features of market microstructure, emergent properties of investor behaviour, artefacts of modelling choices, or merely retrospective labels imposed on continuous processes? As documented in Section III, the corpus reveals near-universal absence of explicit, operational regime definitions, with papers operating under implicit and mutually contradictory assumptions about regime ontology.

The Epistemological Question: How can regimes be reliably identified? Different methods embody incompatible epistemological commitments, not merely technical ones. Frequentist approaches treat regimes as fixed but unknown properties of the data-generating process. Bayesian approaches [8], [9], [71] treat regimes as random variables reflecting the analyst’s beliefs about market structure, updated via Bayes’ rule. These are not interchangeable; papers conflating these interpretations commit category errors that compound downstream.

The Pragmatic Question: Do detected regimes provide economic value? Even if regimes can be statistically identified, do they improve investment decisions, risk management, or forecasting compared to simpler alternatives? Our analysis reveals systematic disconnects between statistical significance and economic utility, with most papers failing to validate practical value [72], [73].

D. Roadmap

This survey makes the following contributions. First, we taxonomise 98 papers across five paradigm families, evaluating each family’s core principles, strengths, and structural limitations (Section II). Second, we document distinct regime conceptualisations and the consequences of definitional chaos (Section III). Third, we conduct a meta-analysis of validation practices, introducing the detection–forecasting distinction and the artefact hypothesis (Section IV). Fourth, we document implementation realities that render most academic strategies impractical (Section V). Fifth, we explain market-specific heterogeneity and present a comparative case study (Section VI). Sixth, we propose reproducibility standards (Section VII).

II. A CRITICAL TAXONOMY OF REGIME DETECTION APPROACHES

The regime detection literature spans diverse methodological traditions, each embodying different assumptions about

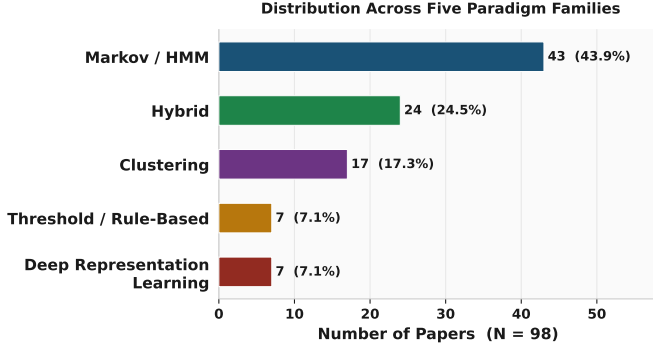


Fig. 2: Distribution of 98 papers across five paradigm families. Markov/HMM methods dominate at 43.9%; Hybrid is a meta-category counted exclusively here. Threshold/Rule-Based and Deep Learning each account for 7.1%.

market structure and optimal detection strategies. We organise 98 papers into five major paradigm families in order of prevalence, analysing their core principles, representative implementations, and fundamental limitations (Figure 2). Table I summarises key statistics for each paradigm family.

A. Markov-Switching and Hidden Markov Models

Markov-switching models (MSMs) and Hidden Markov Models (HMMs) dominate the literature with 43 papers (43.9%). These approaches model regimes as discrete latent states governing observable market dynamics.

Core Framework. Let $\mathbf{y}_{1:T} = (y_1, \dots, y_T)$ denote the observed return series and $\mathbf{s}_{1:T} = (s_1, \dots, s_T)$ the corresponding sequence of unobserved regime states, where $s_t \in \{1, \dots, K\}$. An HMM assumes the following generative structure:

- 1) *Initial distribution:* $\pi_k = P(s_1 = k)$, for $k = 1, \dots, K$.
- 2) *Transition probabilities:* the state sequence follows a first-order Markov chain with time-homogeneous transition matrix $\mathbf{P} \in [0, 1]^{K \times K}$, where $P_{jk} = P(s_{t+1} = k | s_t = j)$ and $\sum_{k=1}^K P_{jk} = 1$.
- 3) *Emission distributions:* conditional on the current state, observations are drawn independently from a regime-specific density, $y_t | s_t = k \sim p_k(y_t | \theta_k)$, where θ_k collects the regime- k parameters.

In the Gaussian case, which is the specification adopted by the majority of surveyed papers, including [47], [7], and [69], the emission density is $p_k(y_t) = \mathcal{N}(y_t; \mu_k, \sigma_k^2)$, so that $\theta_k = (\mu_k, \sigma_k^2)$. The complete-data log-likelihood factorises as

$$\ell(\Theta) = \sum_{t=1}^T \log p(y_t | s_t; \theta_{s_t}) + \sum_{t=2}^T \log P_{s_{t-1}, s_t} + \log \pi_{s_1}, \quad (1)$$

where $\Theta = \{\pi, \mathbf{P}, \{\theta_k\}_{k=1}^K\}$. Because $\mathbf{s}_{1:T}$ is latent, direct maximisation of (1) is intractable. Parameters are instead estimated via the Expectation–Maximisation (EM) algo-

rithm (Baum-Welch procedure), whose E-step computes the smoothed state probabilities

$$\gamma_t(k) = P(s_t = k | \mathbf{y}_{1:T}; \Theta^{(\text{old})}) \quad (2)$$

through recursive forward-backward message passing, and whose M-step yields closed-form updates:

$$\hat{\mu}_k = \frac{\sum_t \gamma_t(k) y_t}{\sum_t \gamma_t(k)}, \quad \hat{P}_{jk} = \frac{\sum_{t=2}^T \xi_t(j, k)}{\sum_{t=2}^T \gamma_{t-1}(j)}, \quad (3)$$

where $\xi_t(j, k) = P(s_{t-1} = j, s_t = k | \mathbf{y}_{1:T})$ is the two-slice smoothed marginal [69]. Once converged, the most probable state sequence is decoded via the Viterbi algorithm:

$$\hat{s}_{1:T} = \arg \max_{\mathbf{s}_{1:T}} P(\mathbf{s}_{1:T} | \mathbf{y}_{1:T}; \hat{\Theta}). \quad (4)$$

GARCH Extensions. A structurally important extension replaces the constant within-regime variance σ_k^2 with a regime-dependent GARCH process. In the Markov-Switching GARCH (MS-GARCH) specification adopted by [74] and implemented in the MSGARCH package [74], the conditional variance in state k evolves as

$$h_{t,k} = \omega_k + \alpha_k \varepsilon_{t-1}^2 + \beta_k h_{t-1,k}, \quad \varepsilon_t = \sqrt{h_{t,s_t}} z_t, \quad z_t \sim \mathcal{N}(0, 1). \quad (5)$$

Maximum-likelihood estimation of the multivariate Student- t variant of this family is developed in [75], while the frequency of GARCH-regime switching is studied empirically in [14]. Multivariate extensions, employed by [76] and the Bayesian nonparametric panel specification of [77], replace the scalar variance with a $d \times d$ regime-dependent covariance matrix Σ_k , so that $\mathbf{y}_t | s_t = k \sim \mathcal{N}_d(\mu_k, \Sigma_k)$. [77] further extend this framework to panel data by placing a hierarchical Pitman-Yor process prior over the regime-specific parameter vectors, thereby allowing the number of cross-sectional clusters within each regime to be learned from the data rather than fixed a priori.

Time-Varying Transitions. The standard assumption that $\mathbf{P}$ is time-invariant is relaxed by [78], who condition transition probabilities on a vector of observable covariates $\mathbf{x}_t$ via

$$P_{jk,t} = \frac{\exp(\mathbf{x}_t^\top \boldsymbol{\lambda}_{jk})}{\sum_{\ell=1}^K \exp(\mathbf{x}_t^\top \boldsymbol{\lambda}_{j\ell})}, \quad (6)$$

which is a multinomial logit applied column-wise to the transition matrix. This modification allows the probability of regime switches to depend on macroeconomic indicators and sentiment variables, addressing one of the most frequently cited limitations of the standard MSM.

Representative Applications. [7] pioneer HMM application to bull/bear market identification using regime-dependent means and volatilities. [74] develop Bayesian MS-GARCH for volatility forecasting. [9] integrate macroeconomic indicators into regime transition dynamics via Eq. (6). [11] and [79] provide comparative assessments of HMM state selection in equity and multi-asset contexts respectively. [10] and [12] extend the Markov autoregressive framework to African emerging markets, demonstrating regime heterogeneity across geographic contexts. [16] introduce a segmented generalisation relaxing the first-order Markov property. [17]

develop a variable-order regime-switching model that selects memory length from data. Recent work extends to panel data [77] and cryptocurrency markets [47], where the four-state model is selected by minimising $AIC = -9021.57$ and $BIC = -8862.20$.

Critical Limitations. Three structural assumptions limit generalisability. *First*, time-invariant transition probabilities: $\mathbf{P}$ is assumed constant, but market micro-structure evolves [47]. *Second*, a fixed number of regimes: researchers must specify K ex ante, most commonly via AIC or BIC evaluated on the very data used for estimation, which is a form of circular model selection. *Third*, conditional independence: observations are assumed independent given the state, so the within-regime autocorrelation structure is fully absorbed by θ_k , which forces the GARCH specification in Eq. (5) to carry the entire burden of temporal dependence. These restrictions force regime structure into parametric templates that may not reflect actual market dynamics.

B. Hybrid Approaches: Combining Multiple Paradigms

Hybrid methods (24 papers, 24.5%) combine techniques from multiple paradigm families, attempting to leverage complementary strengths while mitigating individual weaknesses.

Design Philosophy. Hybrid approaches typically decompose the detection problem into a *labelling stage* and a *prediction stage*, linking the two via a supervised or reinforcement-learning model. Common architectural patterns include: (1) rule-based regime definition with machine-learning prediction [3]; (2) clustering for labelling followed by supervised classification [18]; (3) HMM detection combined with deep-learning forecasting [20]; and (4) signal processing for segmentation with statistical modelling of transitions [19].

Formal Structure. Let the labelling function be $\mathcal{L} : \mathcal{Y} \rightarrow \{1, \dots, K\}$, mapping the observed series to regime labels. The prediction model $f_\phi : \mathcal{X} \rightarrow \{1, \dots, K\}$, parameterised by ϕ , is then trained to minimise the empirical risk

$$\hat{\phi} = \arg \min_{\phi} \frac{1}{T} \sum_{t=1}^T \mathcal{L}_{\text{pred}}(f_\phi(\mathbf{x}_t), \mathcal{L}(\mathbf{y}_{1:t})), \quad (7)$$

where $\mathbf{x}_t$ is a feature vector and $\mathcal{L}_{\text{pred}}$ is a task-specific loss (e.g. cross-entropy for classification). In [3], $\mathcal{L}$ is a peak-to-trough threshold rule ($\pm 15\%$) and f_ϕ is an ensemble of XGBoost and BaggingClassifier models whose outputs are integrated via a hybrid HMM–ensemble voting scheme; in [20], $\mathcal{L}$ is an HMM decoder [Eq. (4)] and f_ϕ is an LSTM network whose output is consumed by a reinforcement-learning trading agent.

Representative Implementations. [3] apply a $\pm 15\%$ peak-to-trough oracle labelling rule to define three regimes (bull, bear, neutral) with a seven-day neutral buffer, then train XGBoost and BaggingClassifier ensembles integrated via a hybrid HMM–ensemble voting framework for regime shift detection and trading. [20] combine HMM regime identification with LSTM forecasting, applying Proximal Policy Optimization for trading decisions. [19] employ quantum-classical support vector machines for regime classification in emerging markets. [21] combine gradient-boosted trees (GBDT) with Shapley

values for interpretable regime change detection, while [22] apply ensemble classifiers to macroeconomic regime identification. [23] develop a hybrid causal-regime framework showing that explicit regime conditioning improves causal inference in non-stationary financial time series. [24] and [25] apply hybrid ML pipelines to equity market regime analysis, documenting performance gains from regime-aware feature engineering. [26] leverage deep reinforcement learning in a hybrid regime-switching investment context.

Complexity-Performance Trade-off. Hybrid methods add substantial complexity, including multiple components, additional hyperparameters, and integration challenges, without consistent performance gains. When components embody contradictory assumptions (discrete states in HMM versus continuous representations in neural networks), the resulting framework lacks theoretical coherence. Moreover, hybrid complexity exacerbates overfitting risk and reduces interpretability without commensurate benefits in generalisation. The fundamental issue is that the labelling stage $\mathcal{L}$ embeds its own assumptions about regime structure, which are then “baked in” to the training targets of f_ϕ ; errors in $\mathcal{L}$ compound through the pipeline with no principled mechanism for correction.

C. Clustering and Unsupervised Learning

Clustering approaches (17 papers, 17.3%) treat regime detection as unsupervised pattern discovery, partitioning market states based on feature similarity without predetermined structure.

K-Means and Mixture Models. Given a feature matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T]^\top$, K -means assigns each observation to the nearest centroid $\{\mathbf{c}_k\}_{k=1}^K$ by solving

$$\min_{\{\mathbf{s}_t\}, \{\mathbf{c}_k\}} \sum_{t=1}^T \sum_{k=1}^K \mathbf{1}[s_t = k] \|\mathbf{x}_t - \mathbf{c}_k\|^2. \quad (8)$$

Gaussian Mixture Models (GMMs) embed this as a probabilistic variant by replacing the hard indicator with soft responsibilities $r_{tk} = P(s_t = k \mid \mathbf{x}_t)$, estimating μ_k , Σ_k , and mixing weights π_k via EM [80]. When returns exhibit heavy tails or non-Gaussian dependence, covariance-based methods cluster on *realised covariance matrices* $\hat{\Sigma}_t \in \mathcal{S}_{++}^d$ (the manifold of $d \times d$ symmetric positive definite matrices) using Riemannian metrics. For the log-Euclidean metric, the geodesic distance is

$$d_{\text{LE}}(\mathbf{A}, \mathbf{B}) = \|\log(\mathbf{A}) - \log(\mathbf{B})\|_F, \quad (9)$$

where $\log(\cdot)$ denotes the matrix logarithm [30].

Statistical Jump Models. A methodologically distinct approach is the *statistical jump model* (JM) of [5] and [29], which replaces the distance-only objective of (8) with a regularised formulation that penalises regime transitions:

$$\min_{\{s_t\} \in \{1, \dots, K\}^T} \sum_{t=1}^T \|\mathbf{x}_t - \mathbf{c}_{s_t}\|^2 + \lambda \sum_{t=2}^T \mathbf{1}[s_t \neq s_{t-1}]. \quad (10)$$

where $\lambda > 0$ is a jump penalty controlling the trade-off between cluster fit and regime persistence. The global

optimum of (10) is obtained exactly via dynamic programming in $O(KT)$ time. A continuous relaxation replaces the discrete indicator $\mathbf{1}[s_t \neq s_{t-1}]$ with a squared ℓ_1 penalty on consecutive soft-assignment vectors, yielding the *fuzzy jump model* of [81], in which regime memberships vary on the probability simplex. [28] employ path signatures up to depth four as features $\mathbf{x}_t$, with cluster counts selected via the maximal eigengap of the affinity matrix. Hierarchical variants build nested regime structures [82], while Wasserstein K -means and Maximum Mean Discrepancy K -means [83] operate directly on distributional representations of rolling windows, bypassing the need for summary statistics. [31] construct correlation networks and apply community detection algorithms as an alternative clustering paradigm for equity regime identification. Information-theoretic clustering using entropy-based flow measures is explored in [32], while [33] develop Hilbert-space nonparametric segmentation that avoids distributional assumptions entirely.

Strengths and Limitations. Clustering's primary advantage is freedom from parametric assumptions about regime structure; algorithms discover patterns directly from data. However, this flexibility creates interpretability challenges. Identified clusters may reflect dominant patterns in training data rather than stable regime structures. Without temporal modelling, vanilla clustering methods struggle with regime persistence and transition dynamics, a limitation that the jump model formulation of (10) partially addresses through the penalty λ . Most critically, clusters lack economic interpretation: a high-volatility cluster may not correspond to an actionable or consistently defined market state across different estimation windows.

D. Threshold and Rule-Based Methods

Threshold and rule-based approaches (7 papers, 7.1%) define regimes via explicit decision rules applied to observable market indicators.

Common Implementations. Peak-to-trough drawdown thresholds signal bull/bear/neutral regimes [3]. Volatility thresholds partition high/low volatility states [35]. Relative extrema in price series define regime boundaries [34]. Macroeconomic indicator rules classify expansion/recession states [9]. Technical analysis patterns trigger regime labels [84]. [36] apply predictability-based threshold tests to the US equity premium, identifying persistent regimes of in-sample predictability. [37] extend threshold logistic models to bear/bull classification with probit dynamics. [38] propose a hybrid threshold-Markov framework for forecasting market states from graph-theoretic features. The directional change framework of [85] and the related approach of [79] define regime transitions endogenously from price extrema rather than imposing external calendar thresholds.

In formal terms, a single-indicator threshold rule takes the form

$$s_t = k \iff \tau_{k-1} < m_t \leq \tau_k, \quad k = 1, \dots, K, \quad (11)$$

where m_t is the indicator value at time t and $-\infty = \tau_0 < \tau_1 < \dots < \tau_K = +\infty$ are fixed thresholds. For the VIX-based

three-regime classifier of [86], $m_t = \text{VIX}_t$, $K = 3$, and τ_1, τ_2 are set to the 33rd and 67th rolling percentiles of the previous 252 trading days. Multi-indicator rules generalise (11) to

$$s_t = g(\mathbf{m}_t; \boldsymbol{\tau}), \quad (12)$$

where $g : \mathbb{R}^p \rightarrow \{1, \dots, K\}$ is a deterministic assignment function and $\mathbf{m}_t \in \mathbb{R}^p$ is a vector of indicators. [87] decompose financial turbulence via the Mahalanobis distance

$$d_t^2 = (\mathbf{y}_t - \bar{\boldsymbol{\mu}})^\top \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{y}_t - \bar{\boldsymbol{\mu}}) \quad (13)$$

into orthogonal *magnitude surprise* and *correlation surprise* components, then apply a two-dimensional version of (12) to classify four regime types.

Transparency versus Sophistication Trade-off. Rule-based methods offer maximum interpretability and computational simplicity. Practitioners can understand and replicate regime assignments without advanced statistical infrastructure. However, this transparency comes at the cost of sophistication. Rules reflect researcher judgment about relevant indicators and appropriate thresholds, introducing subjective biases. Performance depends critically on rule specification, with no principled method for selecting optimal $\boldsymbol{\tau}$ from the infinite space of possibilities. Moreover, discrete thresholds in (11) impose artificial discontinuities on continuous market dynamics: a one-unit change in m_t near τ_k induces a full regime switch, while an identical change far from any threshold has no effect.

The Sampling Problem. Threshold and rule-based methods, like all paradigms surveyed here, operate exclusively on time-based bars (daily, hourly, or minute-level returns). The theoretical case for information-based sampling schemes (dollar bars, volume bars, and tick bars) as a preferable alternative is developed in Section III-D; it applies to all paradigms but is particularly acute for threshold methods whose trigger points are calibrated to time-based indicator values that change character under alternative sampling.

E. Deep Representation Learning

Deep learning approaches (7 papers, 7.1%) employ neural architectures to learn latent regime representations optimised for prediction or decision-making objectives.

Autoencoder-Based Representations. An autoencoder learns a compressed representation by training an encoder $f_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^m$ (with $m \ll d$) and a decoder $g_\psi : \mathbb{R}^m \rightarrow \mathbb{R}^d$ to minimise reconstruction error:

$$\min_{\phi, \psi} \frac{1}{T} \sum_{t=1}^T \|g_\psi(f_\phi(\mathbf{y}_t)) - \mathbf{y}_t\|^2. \quad (14)$$

[41] and [42] extend this to SPD correlation matrices, encoding them via manifold-aware BiMap layers that map $\boldsymbol{\Sigma}_t \mapsto \mathbf{W}\boldsymbol{\Sigma}_t\mathbf{W}^\top$ (preserving positive definiteness), followed by ReEig and LogEig operations:

$$\begin{aligned} \text{ReEig: } \boldsymbol{\Sigma} &\mapsto \mathbf{U} \text{diag}(\max(\boldsymbol{\lambda}, \epsilon)) \mathbf{U}^\top; \\ \text{LogEig: } \boldsymbol{\Sigma} &\mapsto \mathbf{U} \log(\boldsymbol{\lambda}) \mathbf{U}^\top, \end{aligned} \quad (15)$$

where $\boldsymbol{\Sigma} = \mathbf{U} \text{diag}(\boldsymbol{\lambda}) \mathbf{U}^\top$ is the eigendecomposition.

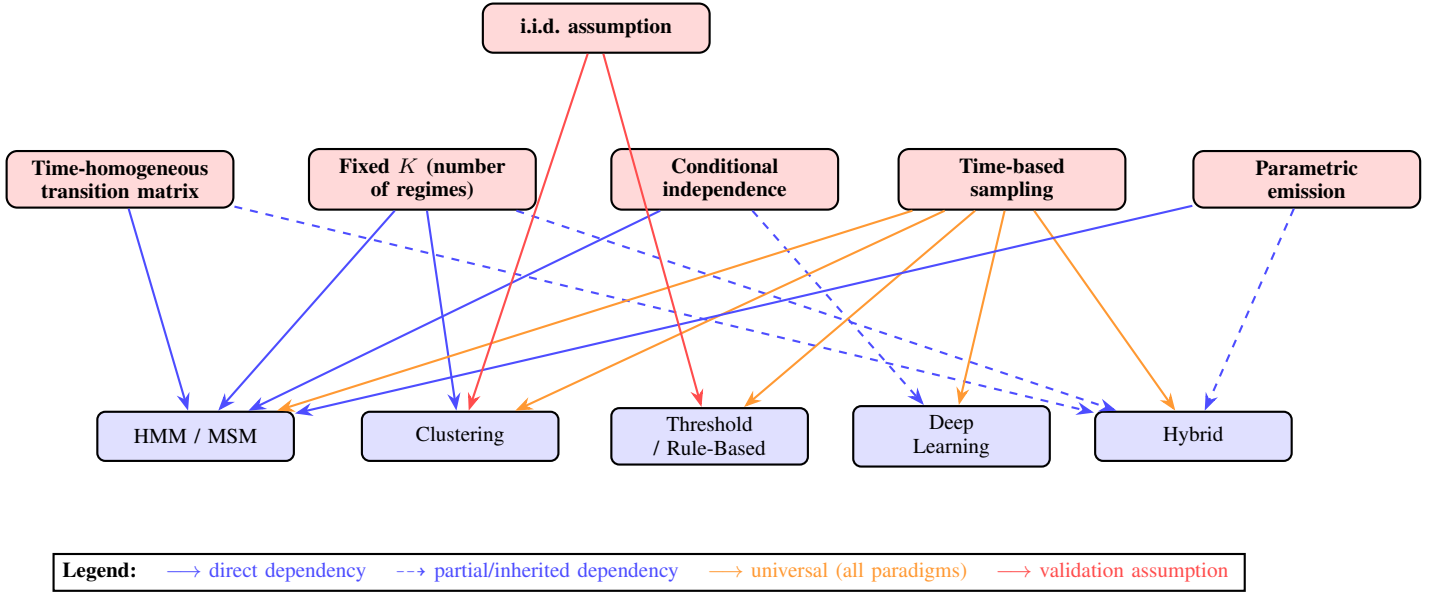


Fig. 3: Assumption dependency graph for the five regime detection paradigms. Red boxes denote foundational assumptions; blue boxes denote paradigm families. Directed edges indicate that a method relies (directly or partially) on the linked assumption. Every paradigm inherits time-based sampling as a universal assumption, yet this is never explicitly validated in any surveyed paper. Fragmentation is driven by incompatible foundational assumptions, not merely by algorithmic differences: HMMs require time-homogeneous transition matrices, fixed K , conditional independence, and parametric emissions simultaneously, while clustering requires only fixed K and implicitly i.i.d. feature distributions; these requirements cannot be simultaneously relaxed and fulfilled by any single framework.

Recurrent and Mixture-Density Networks. LSTMs capture temporal dependencies via gated hidden states:

$$\begin{aligned} \mathbf{h}_t &= \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \\ \mathbf{c}_t &= \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \end{aligned} \quad (16)$$

where $\mathbf{f}_t$, $\mathbf{i}_t$, $\mathbf{o}_t$ are the forget, input, and output gates, respectively, and $\odot$ denotes the Hadamard product [40]. Mixture Density Networks (MDNs) combine an LSTM encoder with a mixture-model output head, emitting regime probabilities as a weighted sum of component densities:

$$p(y_{t+1} | \mathbf{h}_t) = \sum_{k=1}^K \pi_k(\mathbf{h}_t) \mathcal{N}(y_{t+1}; \mu_k(\mathbf{h}_t), \sigma_k^2(\mathbf{h}_t)), \quad (17)$$

where π_k , μ_k , σ_k are functions of the LSTM hidden state [39].

Transformer and Hypernetwork Architectures. Transformers compute pairwise attention scores across the sequence,

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}, \quad (18)$$

enabling long-range regime identification without the recurrence bottleneck [71]. [39] further introduce a *Hypernetwork*: a secondary network $H_\eta : \mathbb{R}^q \rightarrow \mathbb{R}^{|\phi|}$ that generates the parameters ϕ of the primary forecasting model as a function of current regime-indicator inputs $\mathbf{z}_t$ (VIX, rolling volatility, momentum):

$$\phi_t = H_\eta(\mathbf{z}_t), \quad \hat{y}_{t+1} = f_{\phi_t}(\mathbf{y}_{t-L+1:t}), \quad (19)$$

achieving a reported regime-detection accuracy of 95% with an average lag of 2.3 days on the test set. Neural architecture

search for regime-aware model selection is explored in [43], which proposes differentiable architecture search constrained to regime-consistent weight-sharing.

Variational Autoencoders and Regime Identification. [88] develop a Mixture Variational Autoencoder (Mixture-VAE) in which the encoder infers a posterior $q_\phi(\mathbf{z}_t | \mathbf{y}_t)$ over a latent code $\mathbf{z}_t$ belonging to one of K Gaussian components, trained by maximising the evidence lower bound (ELBO):

$$\begin{aligned} \mathcal{L}_{\text{ELBO}} &= \mathbb{E}_{q_\phi(\mathbf{z} | \mathbf{y})} [\log p_\psi(\mathbf{y} | \mathbf{z})] \\ &\quad - D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{y}) \| p(\mathbf{z})). \end{aligned} \quad (20)$$

The authors establish a theoretical link between Mixture-VAE and statistical jump models, showing that (20) with a compound-symmetric transition matrix recovers the jump model objective of (10) as a special case.

Representation Learning Paradigm. Unlike traditional approaches that detect predefined regimes, deep learning discovers latent representations optimised for downstream tasks. This task-driven philosophy shifts focus from regime identification to predictive performance. However, learned representations sacrifice interpretability: neural networks operate as black boxes, providing predictions without explaining what regime structure was discovered or why it matters.

Data Hunger and Overfitting. Deep learning's data requirements create challenges for financial applications. Training requires extensive historical data spanning multiple regime cycles, yet financial data suffers from nonstationarity. Old observations may reflect obsolete market structure. This tension encourages overfitting to recent patterns that may not persist. Moreover, deep architectures amplify hyperparameter

sensitivity, with performance highly dependent on architecture choices, activation functions, and training procedures that resist principled selection.

F. The Fragmentation Problem

This taxonomic analysis exposes fundamental fragmentation. Methods make incompatible assumptions about regime structure (discrete states versus continuous variations), detection strategies (model-based inference versus data-driven discovery versus rule-based classification), and optimisation objectives (likelihood maximisation in (1), cluster-fit minimisation in (8)–(10), or reconstruction loss in (14)). Consequently, papers cannot be meaningfully compared: an HMM finding three regimes and a clustering approach identifying five clusters are incommensurable results, even when applied to identical data.

More troubling, each paradigm embodies untested assumptions (Figure 3). Markov models assume time-invariant transition matrix $\mathbf{P}$. Clustering assumes stable feature-space partitions across estimation windows. Deep learning assumes the training distribution is representative of future dynamics. Rule-based methods assume researcher-chosen thresholds τ capture the relevant regime-generating process. None of these assumptions has been rigorously validated against a common benchmark, yet each methodology proceeds as if its foundational premises were self-evident. Until the field confronts these assumption conflicts and develops methods for adjudicating between approaches, convergence remains impossible (Figure 4).

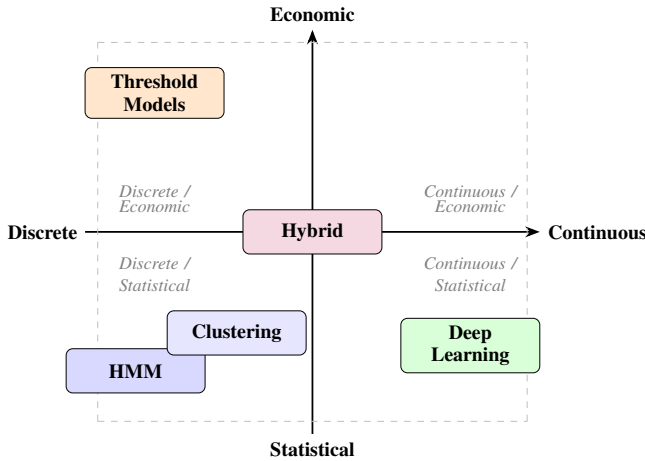


Fig. 4: Conceptual map of regime ontologies across two dimensions: the Discrete–Continuous axis (how regime boundaries are represented) and the Statistical–Economic axis (the theoretical grounding of regime definitions). The spatial separation of methods illustrates why methodological convergence is structurally impossible: HMMs, Hybrid, clustering, threshold models, deep learning, occupy incompatible ontological positions. This is not merely a difference in algorithms but a difference in the fundamental nature of the object being detected. No single method can occupy all quadrants simultaneously, so paradigm unification without ontological commitment is categorically infeasible.

G. The Practitioner-Academic Gap

The 98-paper corpus surveyed here represents exclusively academic publications. This creates a systematic selection bias: the methods studied are not necessarily those used by the practitioners whose decisions regime detection is meant to inform. Available evidence from practitioner literature and institutional investment frameworks suggests a meaningful divergence between the two communities, though systematic data on this gap is limited and should be treated cautiously. Practitioner-oriented work tends to emphasise observable macro-state definitions (growth, inflation, liquidity, and tail-risk conditions) as the organising framework for regime analysis, whereas the academic literature concentrates on latent statistical states estimated via HMMs or deep learning [72], [73]. Regime signals anchored to interpretable threshold indicators, such as VIX-based stress measures, yield curve slope, or purchasing managers indices, feature more prominently in practitioner documentation than in the academic papers surveyed here [6], [73]. Whether this reflects a genuine preference for interpretability, data infrastructure constraints, or simply different institutional mandates is not fully clear from the available evidence. What does appear consistent is that purely statistical clustering approaches without accompanying economic narrative tend to be less prominent in practitioner-facing literature, possibly because a cluster that cannot be linked to an investment decision offers limited operational value [72]. These observations suggest a gap worth investigating empirically, but the academic–practitioner divide in regime detection should not be overstated given the limited systematic evidence available.

This tentative gap is nonetheless visible in the validation data: deep learning and hybrid approaches achieve the highest robustness rates in academic papers (85.7% and 62.5% respectively) yet have the lowest documented adoption in practitioner literature. Whether complexity that improves academic metrics simultaneously reduces economic interpretability in ways that matter for practitioners remains an open empirical question.

III. WHAT ARE REGIMES? DEFINITIONAL CHAOS AND CONCEPTUAL FRAGMENTATION

A scientific field requires shared definitions of its core concepts. Regime detection fails this basic test. This section documents the nature and consequences of definitional fragmentation across the 98-paper corpus. Table III classifies all 98 papers by ontology type.

A. The Absence of Explicit Definitions

Only 1 of 98 papers provides an explicit regime definition with operational classification rules [89]. This sole exception defines regimes based on macroeconomic phases using the OECD G20 Composite Leading Indicator (CLI), categorising market conditions as expansion, recovery, slowdown, and contraction based on the CLI's level relative to 100 and its monthly change direction. While this provides replicable classification for macro portfolio allocation, it exhibits three critical limitations. First, it is **application-specific**. The four-phase framework cannot transfer to cryptocurrency markets,

Paradigm	Papers	%	Robustness (%)	Key Strength	Key Limitation
Markov/HMM	43	43.9	53.5	Formal probabilistic framework	Time-invariant transition matrix
Hybrid	24	24.5	62.5	Component flexibility	Added complexity without consistent gains
Clustering	17	17.3	76.5	No parametric assumptions	Weak temporal structure
Threshold/Rule	7	7.1	71.4	Transparency and simplicity	Arbitrary threshold selection
Deep Learning	7	7.1	85.7	Flexible representation	Black-box; overfitting risk
Total	98	100	63.3		

TABLE I: Taxonomy of Regime Detection Approaches: Summary Statistics. Robustness rate = proportion of papers in that paradigm reporting at least one of: sensitivity analysis (parameter variation tests) or cross-market/cross-asset validation; a paper performing both is counted once. Rates are within-paradigm percentages and therefore need not sum to any fixed total across paradigms.

Notes: See text for robustness rate definition. Coding followed a predetermined rubric; inter-rater reliability was not formally assessed.

high-frequency trading, or intraday regimes where macroeconomic indicators are irrelevant. Second, it is **operationally defined without conceptual foundation**. It specifies *how* to classify observations (CLI thresholds and change direction) but not *what* properties make something a regime in principle. Third, it is **context-dependent**. It requires OECD data access and assumes the four-phase structure reflects universal market dynamics.

The remaining papers either provide no definition whatsoever or rely on implicit assumptions embedded in methodology choices. This creates interpretative chaos. When [7] detect two HMM states with different return means and volatilities, are these the same regimes as the volatility clusters identified by [30] or the trend patterns labelled by [3]? Without explicit definitions specifying what properties constitute a regime, we cannot tell.

The few papers attempting definitions reveal deep conceptual disagreements. Some define regimes by statistical properties: “Persistent periods with stable conditional distributions of returns” [1], [2]. Others emphasise behavioural characteristics reflecting distinct investor sentiment or risk appetite [9], [32]. Still others focus on market microstructure liquidity states [49], [68]. These are not equivalent definitions. Statistical stability does not imply behavioural homogeneity. Behavioural phases need not exhibit microstructural persistence. The field treats fundamentally different phenomena as interchangeable instantiations of generic “regimes.”

B. Seven Dominant Conceptualisations

Analysing implicit assumptions embedded in methodologies and applications reveals seven dominant conceptualisations, each incompatible with the others (see also Table III):

(1) **Trend Regimes:** Bull markets, bear markets, and sideways periods defined by sustained directional price movements. This conceptualisation dominates practitioner applications and rule-based methods [36], [37], [90]. However, what constitutes “sustained”? One month? One year? Multiple papers use identical terminology (bull/bear) for patterns operating at radically different time scales.

(2) **Volatility Regimes:** High volatility versus low volatility states, often associated with crisis periods versus normal conditions. This conceptualisation underlies many GARCH-based approaches [14], [74], [75]. But volatility alone cannot

define regimes. High volatility occurs in both sustained downturns and sharp rebounds, which require different investment strategies despite similar volatility profiles.

(3) **Correlation Regimes:** Periods of high versus low cross-asset correlation, relevant for portfolio diversification. This conceptualisation appears in multivariate HMMs and covariance clustering methods [30], [31], [91]. However, correlation regimes may be orthogonal to return regimes. Assets can be highly correlated during both bull and bear markets.

(4) **Macroeconomic Regimes:** Economic expansion, recession, stagflation, or recovery phases. This conceptualisation connects markets to real economic conditions [4], [65], [67] but operates at time scales (quarters to years) incompatible with high-frequency trading applications. Papers equivocate between macroeconomic regimes and market regimes without addressing this time scale mismatch.

(5) **Liquidity Regimes:** Market microstructure states reflecting order flow dynamics, bid-ask spreads, and depth [32], [49], [68]. This conceptualisation emerges in high-frequency applications but rarely connects to lower-frequency regime concepts.

(6) **Risk Appetite Regimes:** Periods of high versus low investor risk tolerance, proxied by volatility indices (VIX), credit spreads, or flight-to-quality patterns [6], [87], [92]. This behavioural conceptualisation links regimes to investor psychology but struggles with empirical operationalisation.

(7) **Structural Break Regimes:** Permanent shifts in market dynamics following regulatory changes, technological innovations, or institutional evolution [15], [50], [93], [94]. Unlike cyclical regimes that rotate (bull to bear to bull), structural breaks are one-directional and irreversible.

These conceptualisations are not refinements of a shared core concept but incommensurable frameworks. A trend regime model cannot detect liquidity regimes; a macroeconomic regime classifier will miss high-frequency structural breaks. Papers apply the same term “regime” to phenomena operating at different time scales, driven by different causal mechanisms, and requiring different identification strategies.

C. The Temporal Scale Problem

Regime conceptualisations fragment further across temporal scales. Our analysis identifies five distinct time horizons (Figure 5):

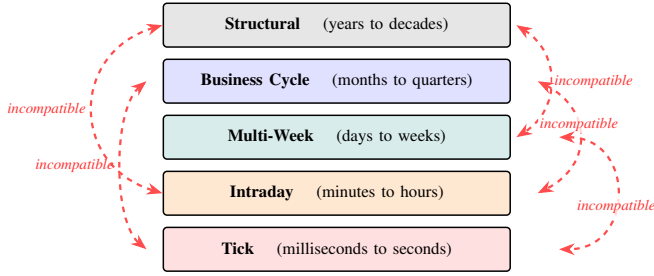


Fig. 5: The five-layer time-scale stack of regime detection. Adjacent layers may share some conceptual overlap, but non-adjacent layers are structurally incompatible: a tick-level microstructure regime cannot inform a structural-change framework, and a business-cycle recession label carries no actionable information for intraday execution. Dashed bi-directional arrows denote cross-scale incompatibilities. Papers rarely acknowledge that regimes detected at one temporal layer are noise at another, and no method in the surveyed literature operates coherently across more than two adjacent layers simultaneously.

Tick-Level Regimes (milliseconds to seconds): Microstructure states governing high-frequency order flow [68], [95]. Relevant for market making and execution algorithms but irrelevant for long-term investors.

Intraday Regimes (minutes to hours): Trading session patterns such as opening volatility and closing auctions [49], [51]. Relevant for day traders but invisible at daily frequency.

Multi-Day Regimes (days to weeks): Short-term trend or volatility patterns. The most common focus in machine learning applications [29], [33], [83].

Business Cycle Regimes (months to quarters): Macroeconomic phases [4], [65], [67]. Relevant for tactical asset allocation but too slow for active trading.

Structural Regimes (years to decades): Long-term market structure changes [2], [15], [94]. Relevant for institutional strategy but offer no actionable signals within typical investment horizons.

Papers rarely acknowledge that regimes detected at one time scale may be noise at another. What appears as persistent high-volatility regime at daily frequency may be rapid transitions between microstructure states at tick frequency, or merely transient fluctuations within a stable macroeconomic expansion at quarterly frequency. Without explicit specification of relevant time scales and validation that detected regimes operate at claimed horizons, regime identification becomes arbitrary pattern labelling.

D. The Sampling Scheme Dimension

Definitional chaos extends to a dimension that even the temporal scale discussion above cannot fully capture: the choice of *sampling scheme*. This choice is not merely technical; it is a further implicit definitional commitment that the literature never acknowledges. [96] demonstrates that time bars (fixed calendar intervals) differ fundamentally from information bars in their statistical properties, and therefore in what “regime structure” they can reveal:

Time Bars: Each bar represents fixed time (1 day, 1 hour, 1 minute). Information arrives irregularly; quiet and active periods are weighted equally; statistical properties are poor (fat tails, autocorrelation).

Volume Bars: Each bar represents fixed share volume traded. Better captures activity but ignores price.

Dollar Bars: Each bar represents fixed dollar volume (price times volume). Normalises for both activity and price; exhibits superior statistical properties [96].

Tick Bars: Each bar represents fixed number of transactions. Captures order flow but ignores trade size.

The consequence for regime detection is direct: a volatility regime apparent at daily frequency might fragment into multiple microstructure states when sampled via dollar bars, or disappear entirely as transient noise. Two papers using identical underlying tick data but different sampling schemes may reach contradictory conclusions about regime structure while both claiming validity. Yet as Table II documents, the current literature exclusively uses time-based sampling (91 of 98 papers), imposing a modelling assumption that has never been explicitly validated and may fundamentally affect reported findings.

Bar Type	Papers (<i>n</i>)	Percentage
Time bars (daily, weekly, monthly, hourly, minute)	91	89.2%
Tick bars (transaction-count bars)	3	2.9%
Volume bars	[not recorded]	
Dollar bars	[not recorded]	
Not applicable / unspecified	8	7.8%

TABLE II: Sampling Scheme Usage across the surveyed papers. Counts reflect how many papers use each bar type as their primary input data.

Notes: The corpus records data frequency (e.g., “Daily,” “1-minute,” “Tick”) but does not contain a dedicated field for volume bars or dollar bars. No paper in the 98-paper empirical corpus uses volume bars or dollar bars as the primary sampling scheme; these values are marked *not recorded in corpus* rather than zero to distinguish absence of evidence from evidence of absence. This finding supports the argument in Section III that the field’s exclusive reliance on time-based sampling is an untested assumption. For the theoretical case for dollar bars, see [96].

E. Consequences of Definitional Chaos

Definitional ambiguity generates three destructive consequences for the field:

Incomparable Results: When papers define regimes differently, their findings cannot be synthesised. One study’s success detecting macroeconomic regimes provides no evidence about microstructure regime detection. Narrative reviews aggregating heterogeneous results as evidence for generic “regime detection” make category errors.

Unfalsifiable Claims: Without explicit definitions specifying what would constitute regime absence, claims that “markets exhibit regime structure” become unfalsifiable. If any detected pattern counts as regime evidence, the hypothesis cannot be tested. Science requires the possibility of negative

Ontology Type	n	Expl. Def.	Dominant Time Scale	Action-ability	Representative Examples
Statistical (latent states, volatility, trend)	73	39 Y / 34 N	Daily/Weekly (52); Intraday (14); Monthly (9)	50 Y / 23 N	[7], [14], [74], [75]
Macro (business-cycle, macro indicators)	22	10 Y / 12 N	Monthly/Qtrly(10); Daily/Weekly (10)	14 Y / 8 N	[4], [9], [65], [67]
Behavioural (risk appetite, sentiment)	2	1 Y / 1 N	Daily/Weekly (2)	1 Y / 1 N	[97]
Structural (permanent market-structure breaks)	1	0 Y / 1 N	Tick (1)	1 Y / 0 N	[68]
Total	98	50 Y / 48 N	—	70 Y / 28 N	—

TABLE III: Regime Definition Matrix grouped by ontology type across all 98 papers. **Expl. Def.** = paper provides an explicit, operational regime definition (Y/N). **Actionability Tested** = paper applies regimes to a trading/allocation strategy (Y/N).

Notes: “Explicit definition” requires an operationalisable rule or formal statement; implicit embeddings in model mechanics are coded N. Time-scale counts reflect the primary data frequency; multi-frequency papers are assigned to their primary frequency. See Figure 4 for the conceptual map of ontology positions and Figure 5 for the five-layer time-scale stack.

findings, but definitional vagueness prevents definitive rejection.

Method Selection Arbitrariness: Without knowing what researchers aim to detect, method choice becomes unprincipled. Should one use HMMs, clustering, or deep learning for regime detection? The answer depends on what regimes are, what properties they exhibit, and what time scales they operate on. Absent consensus definitions, method selection reduces to arbitrary preference or convenience.

IV. THE VALIDATION CRISIS: HOW PAPERS TEST (OR DON’T TEST) THEIR CLAIMS

Scientific claims require rigorous validation. This section exposes systematic deficiencies in how regime detection research tests its findings, revealing widespread practices that inflate reported performance and obscure limitations. Table IV summarises validation practices across all 98 papers.

	Ex-post Detection ($n = 32$ papers)	Ex-ante Forecasting ($n = 69$ papers)
Stat. OOS	4 Ex-post OOS validated	59 Ex-ante OOS validated
Econ. App.	28 Ex-post + econ. test	68 Ex-ante + econ. test

Fig. 6: Detection vs. Forecasting Success Matrix. Each cell shows the number of papers satisfying both the temporal claim (column) and the success criterion (row). Statistical Success = paper reports out-of-sample validation; Economic Success = paper implements and tests a trading or allocation application. The asymmetry between ex-post and ex-ante statistical success (4 vs. 59) reflects the validation crisis: the majority of regime-identification papers do not validate ex-post detections out-of-sample.

A. Out-of-Sample Validation: Essential but Inconsistent

Out-of-sample testing is the minimum standard for time series research. Models should be trained on historical data and evaluated on subsequent periods never seen during fitting. Yet only 63 papers (64.3%) report out-of-sample validation; 35 papers (35.7%) either conduct no temporal split or fail to document evaluation protocols. The asymmetry between statistical and economic success is quantified in Figure 6 and Table V.

Papers employing proper out-of-sample procedures exhibit sobering performance declines. [98] report in-sample Sharpe ratios substantially positive, collapsing to negative values out-of-sample. Dückerhoff [57] acknowledges that strong in-sample regime detection results do not replicate consistently out-of-sample, attributing this to transaction costs and signaling lags under realistic market conditions. These honest assessments highlight the danger of in-sample-only evaluation. Overfitted models perform well on training data precisely because they capture noise as signal.

More concerning, papers without out-of-sample validation often make the strongest claims. This selection effect creates distorted literature where prominently reported successes reflect methodological inadequacy rather than genuine discovery. Until out-of-sample validation becomes mandatory, the field cannot distinguish legitimate regime structure from sophisticated curve-fitting.

B. The Look-Ahead Bias Epidemic

Look-ahead bias occurs when models use information unavailable at prediction time. Our analysis identifies 30 papers (38.0% of the 79 papers making predictive or backtesting claims) with likely or confirmed look-ahead bias. Common violations include:

Full-Sample Parameter Estimation: Estimating model parameters on entire datasets, including future periods to be predicted. This contamination means models already “know” future regime characteristics when making historical predictions.

Future Data for Normalisation: Normalising features using full-sample statistics, leaking future mean and variance information into historical predictions.

Regime Labeling Post-Hoc: Assigning regime labels retrospectively after observing full market cycles, then training

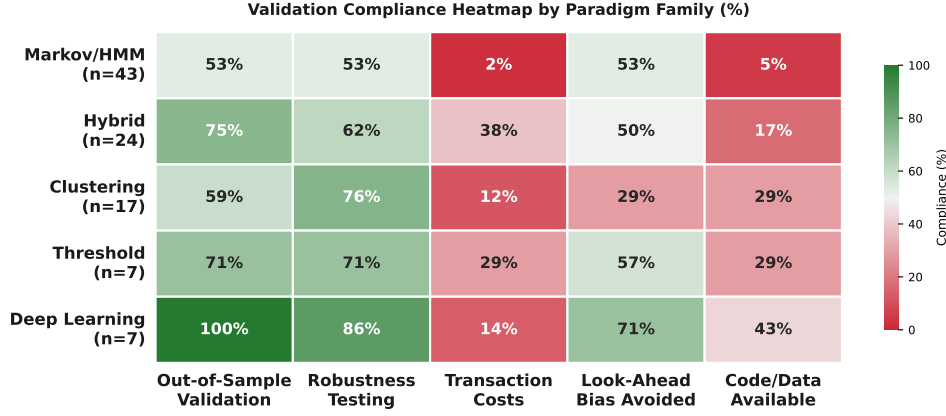


Fig. 7: Validation compliance heatmap by paradigm family. Each cell reports the percentage of papers in that family meeting the criterion. Red indicates low compliance; green indicates high compliance. Transaction costs (column 3) and code/data availability (column 5) are systemically neglected across all families. Deep Learning leads on OOS validation (100%) and robustness (86%) but the family comprises only 7 papers. Markov/HMM, the dominant family ($n = 43$), scores worst on transaction costs (2%) and code availability (5%).

classifiers to predict these labels. Labels themselves incorporate future information.

Full-Period Backtests: Running strategies over entire sample periods including data used for parameter optimisation.

Look-ahead bias generates illusory performance. A model exhibiting 85% regime prediction accuracy with future information may achieve only 55% accuracy in genuine forward prediction. Yet papers rarely distinguish contaminated from clean performance. An additional 19 papers (19.4%) are purely ex-post descriptive studies for which look-ahead bias is not applicable; these make no predictive claims and are appropriately limited to historical analysis.

C. Data Split Protocols: The Devil in Implementation Details

Even papers conducting out-of-sample evaluation often employ flawed data split protocols. Our analysis reveals concerning patterns: inadequate holdout periods (many papers reserve only 10–20% of data for testing), absence of temporal cross-validation, optimisation on test sets, and cherry-picked evaluation periods. These implementation details matter enormously. A model validated on carefully selected holdout data may fail catastrophically when deployed on genuinely unseen market conditions. Figure 7 provides a cross-paradigm view of validation compliance.

D. Robustness Testing: Who Actually Tests Their Claims?

Robustness testing assesses whether results depend critically on specific choices. Our analysis reveals only 62 papers (63.3%) conduct any robustness checks, with 36 papers (36.7%) reporting none. Adoption varies dramatically by methodology: Deep learning approaches test robustness 85.7% of the time, clustering methods 76.5%, threshold methods 71.4%, hybrids 62.5%, but Markov/HMM approaches (the field’s most popular paradigm) only 53.5%.

When robustness testing occurs, common approaches include cross-market validation [5], [82], [99], parameter sensitivity [29], [81], [100], temporal robustness across crisis versus

normal conditions [9], [87], and alternative specifications [77], [101].

Papers failing robustness testing often make strong claims without evidence. [47] propose HMM for Bitcoin regime detection but only compare different state numbers on the same dataset. [68] develop asymmetric HMM for microstructure regimes without testing alternative parameterizations. [88] present a methodologically sophisticated Mixture-VAE framework but do not report sensitivity analysis for their central hyperparameter λ .

E. Baseline Comparisons: State of the Literature

Meaningful evaluation requires comparing regime detection against baselines representing the null hypothesis of no regime structure. Despite all 98 papers providing some baseline information, quality varies dramatically. Only 12 papers include buy-and-hold comparisons and only 14 compare against single-regime statistical models, a severe gap, since outperforming other regime models proves only relative superiority within a potentially flawed class, not absolute utility.

When meaningful baselines appear, regime model advantages often shrink dramatically. [102] report their best regime-switching rotation strategy achieves a Sharpe ratio of 1.06 and an annualised return of 22.53%, outperforming equal-weighted benchmarks, yet this study is conducted without transaction costs and the strategy relies on monthly rebalancing, costs that would materially erode reported gains. [98] find all models perform poorly out-of-sample, with the best regime-aware approach achieving a Sharpe ratio of -0.29 versus -1.61 for the no-regime baseline. Both lose money, raising questions about practical utility. Papers that do compare against multiple baseline tiers, including [45], [46], [101], [103]–[105], consistently find that added complexity yields surprisingly modest gains when properly tested. Journals should require at minimum: a naive baseline (buy-and-hold or equal-weight), a single-regime statistical alternative (GARCH, OLS), and explicit net-

of-cost reporting; without this, the literature will continue to systematically overstate the value of regime detection.

V. THE TRANSACTION COST AND EXECUTION REALITY GAP

Academic regime detection research often operates as if detected signals translate instantly into costless trades. Reality imposes harsh constraints through transaction costs, slippage, latency, and market impact that often eliminate theoretical advantages. This section documents the systematic neglect of implementation realities that renders most regime-based strategies impractical.

A. Transaction Costs: The Reality Most Papers Ignore

Every trade incurs bid-ask spreads, exchange fees, taxes, and potentially market impact costs. For regime-based strategies requiring frequent rebalancing as detected states change, these costs compound rapidly. Yet our analysis reveals 83 papers (84.7%) either incorporate no transaction costs or leave them unspecified.

This omission is consequential. Consider a regime strategy generating 15% annual returns before costs. With realistic trading costs of 5–10 basis points per trade and monthly rebalancing, net returns might drop to 8–10% annually, potentially underperforming passive alternatives. A strategy triggering weekly regime-driven trades could see costs exceed 100 basis points annually, eliminating most return advantages.

The papers that do incorporate costs report illuminating results. Portfolio execution under regime switching with non-linear cost functions is modelled in [61], [62]. Optimal execution frameworks accounting for regime-induced slippage are developed in [59]. These honest assessments are rare exceptions. The handful of papers testing regime-based trading net of realistic costs, including [5], [57], [58], [65], [66], [99], [104], [105], consistently find post-cost advantages reduced to modest or negative values. Including realistic costs often eliminates reported profitability, creating publication disincentives for honest accounting.

We propose tiered cost modelling requirements:

Tier 1 (Mandatory for all papers): Apply proportional transaction costs of 10 bps per one-way trade (equities), 2 bps (FX majors), 25 bps (cryptocurrency), or 5 bps (futures). Report both gross and net Sharpe ratios. If net Sharpe ≤ 0 , the paper must explicitly acknowledge non-viability.

Tier 2 (Recommended for strategy papers): Incorporate bid-ask spread estimates from historical data, assume 50% of quoted spread realised, and include 10 bps market impact for trades exceeding 1% of average daily volume.

Tier 3 (Required for high-frequency applications): Model the full execution pipeline: detection latency, order transmission, queue position, fill probability, and adverse selection. At minimum, assume 1-tick slippage for market orders.

B. Slippage and Latency in High-Frequency Applications

High-frequency regime detection faces even harsher execution constraints. Our analysis reveals 97 papers (99.0%) fail

to address slippage or latency despite many using tick-level or minute-level data. In reality, the full execution pipeline introduces detection latency, transmission delays, queue time, market impact, and information leakage. [68] develop asymmetric HMM for tick-level microstructure regime detection but provide no analysis of execution latency or slippage. [95] develop generative-discriminative models for high-frequency financial regimes but similarly omit execution cost analysis, an omission that is particularly problematic given their intraday application. For practical implementation, high-frequency regime models must account for total execution time from signal generation through order completion.

C. Economic Versus Statistical Value: The Fundamental Disconnect

The disconnect between statistical significance and economic utility pervades regime detection literature. Papers report impressive statistical metrics but provide no evidence these translate to profitable trading, better risk management, or superior allocation decisions. Key causes include overfitting, ignored costs, data snooping across multiple model variants [86], [103], [104], and publication bias.

The field would benefit from requiring papers to report not just statistical metrics but economic measures: net returns after all costs, comparison to passive benchmarks on identical evaluation periods, out-of-sample tests on truly held-out data, and realistic implementation assumptions. Until economic validation becomes standard, the disconnect between statistical results and practical value persists.

VI. WHY DIFFERENT MARKETS NEED DIFFERENT REGIME MODELS

A striking finding from our systematic review: regime detection methods rarely generalise across markets. Models trained on equities fail on fixed income; cryptocurrency regime detectors collapse on commodities; high-frequency approaches cannot transfer to monthly macroeconomic data. This section explains why market heterogeneity prevents unified regime detection frameworks.

A. Cryptocurrency versus Equity versus Macro Regimes

Different markets exhibit fundamentally different regime characteristics. Our analysis covers 76 papers studying equities, 15 on cryptocurrencies, 25 on foreign exchange, 16 on commodities, and 10 on multi-asset portfolios (Figure 8).

Cryptocurrency Markets (15 papers): Characterised by extreme volatility, 24/7 trading, heavy retail participation, sensitivity to regulatory news and social media sentiment. Regime changes occur rapidly, sometimes intraday, driven by sentiment shifts rather than fundamental factors [47]–[51]. Traditional Markov models with stable transition probabilities struggle with crypto’s non-stationary dynamics.

Equity Markets (76 papers): Driven by business cycles, earnings expectations, monetary policy, and gradual risk appetite shifts [1], [2], [15], [21], [44]–[46]. Regime changes tend to be slower, measured in months or quarters, correlating with macroeconomic conditions.

Validation Practice	Yes	NoMissing	% Compliant
Out-of-sample testing	63	35	64.3%
Robustness testing	62	36	63.3%
Transaction costs included	15	83	15.3%
Slippage/latency addressed	1	97	1.0%
Baseline comparisons (any)	98	0	100.0%
Buy-and-hold baseline	12	86	12.2%
Single-regime model	14	84	14.3%
Look-ahead bias avoided	49	30	62.0% [†]
Code/data available	16	82	16.3%
All criteria met	0	98	0.0%

TABLE IV: Validation Practices Across 98 Papers.

[†] Among 79 papers making predictive or backtesting claims; 19 additional papers are purely ex-post descriptive and are excluded from this criterion.

Paradigm	<i>n</i>	OOS Test	Tx Costs	Robustness	Code/Data	Mean Score (out of 4*)
Markov/HMM	46	25	3	25	2	1.43
Hybrid	18	14	6	12	4	2.00
Clustering	15	8	2	11	4	1.67
Threshold/Rule	6	4	2	4	2	2.00
Deep Learning	7	7	1	6	3	2.43
Other	6	5	1	4	1	1.83
Total / All	98	63	15	62	16	1.62

TABLE V: Validation Standards Compliance Score by methodological paradigm. Each criterion is scored 1 if met, 0 otherwise, yielding a total score of 0–5. The five criteria are: (1) Out-of-sample test, (2) Transaction costs included, (3) Robustness checks, (4) Baseline comparison, (5) Code/data availability. Baseline comparison is coded 1 if the paper includes any form of baseline beyond other regime methods; all papers scored 1 on this criterion (see Table IV). Only papers coded with a clear Yes or No are scored.

* Score is out of 4 because the “Baseline comparison” criterion is satisfied by all 98 papers and is excluded from the per-paradigm mean to avoid inflating scores. *Score distribution*: median score across all papers = 2 (out of 5 including baseline), range = 1–4; no paper satisfies all five criteria simultaneously.

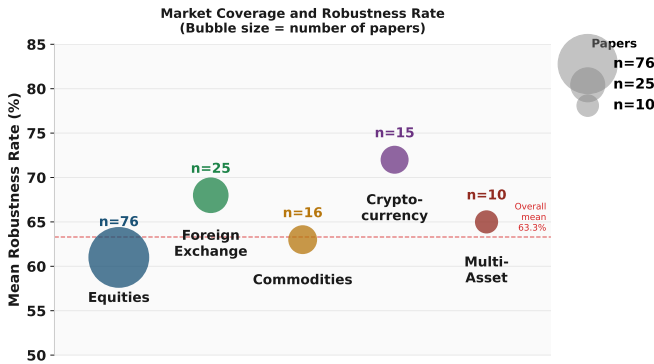


Fig. 8: Market coverage and mean robustness rate by asset class. Bubble size encodes the number of papers. The dashed line marks the overall mean (63.3%). Equities dominate coverage ($n = 76$) yet fall below the mean on robustness, while the smaller cryptocurrency and FX literatures achieve higher validation rates.

Foreign Exchange (25 papers): Influenced by interest rate differentials, trade flows, central bank policy, and geopolitical events [13], [52]–[54]. FX regimes may operate at multiple time scales simultaneously.

Commodities (16 papers): Driven by supply-demand dynamics, inventory levels, weather, and geopolitical shocks [56]. Commodity regimes often exhibit seasonal patterns absent in financial assets.

A model detecting Bitcoin volatility regimes using minute-level data cannot meaningfully transfer to long-term equity allocation. The underlying forces, relevant time scales, and appropriate indicators differ so drastically that “regime detection” becomes a superficial label masking incompatible methodologies.

B. The COVID-19 Stress Test: A Natural Experiment and Future Benchmark

March 2020 represents one of the sharpest and most clearly documented regime transitions in modern financial history. The S&P 500 fell 34% in 33 calendar days, the VIX exceeded 80, and cross-asset correlations collapsed toward unity, a configuration qualitatively distinct from any prior period in the post-2000 data that most regime detection papers use for training. This episode provides a natural experiment: methods trained pre-2020 encountered a genuinely out-of-distribution regime transition.

Of the 98 papers in our corpus, 63 report out-of-sample validation and 68 use data ending in 2020 or later. However,

only a small subset explicitly evaluate regime transition detection performance spanning the March 2020 event. Among the papers that do engage with the COVID-19 period, a consistent pattern emerges: models trained on 2000–2019 data encoded regime structures calibrated to the 2008–2009 crisis, the slowest regime transition in post-2000 data. When March 2020 arrived with a fundamentally different velocity (weeks, not quarters), detection lags were substantial.

We note important limitations here: we were unable to systematically extract comparable quantitative detection lag statistics across papers for the March 2020 period, as most studies did not report them in a standardised format. Notably, 68 papers (69.4%) were published in 2020–2025, after COVID-19, yet only a minority explicitly analyse whether their methodological choices were informed by this episode.

We therefore reframe this section as a proposal rather than a retrospective evaluation: *the February–May 2020 period should become a mandatory standardised stress-test evaluation window for all equity regime detection papers*, analogous to the “financial crisis” evaluation periods used in credit risk modelling. Future surveys should require authors to explicitly report COVID-19 period performance in a standardised format, enabling the first rigorous cross-method comparison on this natural experiment.

C. Case Study: Identical HMM Applied to S&P 500 and Bitcoin

A. Motivation and Design: The following case study operationalises the market heterogeneity argument through a controlled experiment: an identical two-state Gaussian HMM is fitted to S&P 500 and BTC-USD daily log-returns using the same estimation procedure, convergence criterion, and evaluation protocol. Any difference in detection performance and economic value is therefore attributable solely to the structural mismatch between the model and asset-specific regime dynamics, not to methodological heterogeneity.

B. Data: Daily log-returns are constructed from Yahoo Finance adjusted closing prices for the S&P 500 ($\hat{GSPC}$) and BTC-USD, covering 1 January 2018 to 31 December 2023 (approximately 1,510 aligned trading days). The 2018–2023 window contains the BTC bear market of 2018, the COVID-19 crash of March 2020, the synchronised bull run of 2020–2021, the Federal Reserve tightening cycle bear market of 2022, and the FTX collapse of November 2022, providing stress and calm episodes for both assets simultaneously. A temporal split at 31 December 2021 yields 1,007 training days and 501 test days.

C. Model Specification: We fit a two-state Gaussian HMM [69] with initial distribution π , time-homogeneous transition matrix $\mathbf{P} \in [0, 1]^{2 \times 2}$, and Gaussian emissions $y_t \mid s_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$. Parameters are estimated by the Baum-Welch EM algorithm with convergence criterion $|\ell^{(i)} - \ell^{(i-1)}| < 10^{-7}$. Ten random restarts are used and the solution with the highest log-likelihood is retained. States are labelled so that State 0 always carries the lower estimated volatility and State 1 the higher. Strategy evaluation uses exclusively filtered posteriors with a one-bar lag, eliminating all look-ahead bias.

Asset	State	$\hat{\mu}$ (%/yr)	$\hat{\sigma}$ (%/yr)	$\hat{P}_{kk}$	$\mathbb{E}[D_k]$
S&P 500	0 (Lo-Vol)	+36.9	10.4	0.973	37 d
	1 (Hi-Vol)	−61.3	39.8	0.910	11 d
BTC-USD	0 (Lo-Vol)	+38.7	35.4	0.769	4.3 d
	1 (Hi-Vol)	+15.1	110.7	0.692	3.2 d

TABLE VI: Estimated HMM parameters, training period (Jan 2018–Dec 2021, $N = 1,007$ days). State 0 = Low-Vol; State 1 = High-Vol. $\hat{\mu}$, $\hat{\sigma}$ annualised.

$\mathbb{E}[D_k] = (1 - \hat{P}_{kk})^{-1}$. AIC: S&P $-6,455.7$; BTC $-3,445.4$. BTC expected durations of 3–4 days indicate model misspecification rather than genuine regime structure.

Asset	State	Med. (d)	Mean (d)	h_k (d)
S&P 500	0 (Low-Vol)	44.0	37.1	25.4
	1 (High-Vol)	10.0	11.1	7.3
BTC-USD	0 (Low-Vol)	6.5	4.3	2.6
	1 (High-Vol)	3.0	3.2	1.9

TABLE VII: Regime persistence, training period (Jan 2018–Dec 2021). Spell statistics from Viterbi path. $h_k = \ln 2 / (-\ln \hat{P}_{kk})$.

Velocity ratio, S&P Low-Vol vs. BTC Low-Vol: $h_{SP}/h_{BTC} \approx 9.8 \times$. BTC half-lives of 2–3 days fall below any actionable threshold.

D. Estimated Model Parameters: Table VI reports the EM-estimated parameters. The S&P 500 HMM recovers a persistent Low-Vol state ($\hat{P}_{00} = 0.973$, $\mathbb{E}[D_0] \approx 37$ days) and a shorter High-Vol state ($\hat{P}_{11} = 0.910$, $\mathbb{E}[D_1] \approx 11$ days). The annualised volatility ratio $\hat{\sigma}_1/\hat{\sigma}_0 \approx 3.8$ indicates strong distributional separation. The BTC HMM, despite identical estimation code, converges to dramatically less persistent states: $\hat{P}_{00} = 0.769$ and $\hat{P}_{11} = 0.692$, implying expected durations of only 4.3 and 3.2 days respectively. These sub-week expected durations are a diagnostic of model misspecification; the EM algorithm is partitioning BTC’s continuous volatility evolution into two nominal states that switch so rapidly they carry no actionable information.

E. Regime Persistence and Detection Lag: Table VII reports spell statistics. The S&P 500 Low-Vol half-life of 25.4 days versus BTC Low-Vol half-life of 2.6 days represents a near-ten-fold velocity differential. BTC half-lives of 1.9–2.6 days preclude any practical trading response under realistic latency and transaction cost constraints. Detection lag analysis shows BTC encounters 60 transitions over 501 test days (annualised rate 30.2/yr) versus S&P 500’s 12 transitions (6.0/yr), a five-fold difference in transition frequency equivalent to one rebalancing signal every 8.5 trading days for BTC.

F. Out-of-Sample Economic Performance: A regime-aware long-only strategy uses filtered posteriors with a one-bar lag and transaction cost $c = 10$ bps per one-way unit weight change. For S&P 500, the HMM strategy achieves $SR = +0.196$ versus $SR = -0.127$ for buy-and-hold ($\Delta SR = +0.323$), with maximum drawdown reduced from -25.4% to -11.3% . For BTC-USD, the strategy underperforms buy-and-hold ($\Delta SR = -0.190$, $SR = -0.316$ vs. -0.126). The strategy’s excessive rebalancing driven by 60 Viterbi transitions over 501 days erodes returns relative to passive

holding. This is the strongest empirical demonstration of the transition velocity mismatch: not merely reduced effectiveness, but active destruction of value when a slow-regime model is applied to a fast-regime asset.

G. Implications: The S&P 500 HMM partially passes regime-quality criteria; the BTC HMM fails the majority. The scorecard gap arises entirely from the near-ten-fold regime velocity differential between the two markets. Furthermore, BTC achieves 92.4% filtered classification accuracy yet yields negative economic value, while S&P 500 achieves only 88.4% filtered accuracy yet yields positive economic value. Accuracy metrics without reference to the regime’s actionable duration are systematically misleading for high-velocity assets.

VII. THE REPRODUCIBILITY CRISIS AND PATH FORWARD

Science requires reproducibility. Other researchers must be able to verify findings, compare methods on identical data, and build incrementally on prior work. Regime detection fails this standard dramatically.

A. The Data and Code Availability Crisis

Our analysis reveals only 16 papers (16.3%) provide code or data, with 82 papers (83.7%) supplying neither datasets nor implementation code, making replication impossible. Even when data sources are named (“Yahoo Finance,” “Bloomberg”), precise sample construction details (exact tickers, date ranges, adjustment methods, handling of splits and dividends) are typically omitted.

The handful of papers providing code and data enable rare reproducibility successes. [28] share GitHub repositories allowing exact replication. [71] provide detailed implementation specifications. These examples demonstrate feasibility. Publication venues should mandate data and code release as condition for acceptance, with limited exceptions for proprietary data accompanied by detailed synthetic data examples enabling method verification.

B. The Absence of Shared Benchmarks

Computer vision advanced through ImageNet, natural language processing through GLUE benchmarks, but regime detection has no equivalent shared evaluation datasets. Papers test on idiosyncratic samples, preventing method comparison. The field urgently needs standardised benchmarks spanning: multiple asset classes (equities, fixed income, commodities, currencies, cryptocurrencies), different time periods (normal conditions and crises), various frequencies (daily, intraday, monthly data), and geographic diversity (US, European, Asian, and emerging markets [10], [12], [53], [54]).

C. Parameter Estimation Windows: The Hidden Degrees of Freedom

Even papers using identical data can produce incomparable results through different parameter estimation protocols: full-sample estimation (contaminating test periods), expanding windows (avoiding look-ahead bias but introducing parameter instability), or rolling windows (maintaining adaptivity but

with arbitrary length choices). Rather than mandating single protocols, the field should require explicit documentation of estimation sample periods, whether parameters are fixed or updated, retraining frequency, and justification for choices.

D. A Proposal for Unified Standards

Based on analysis of papers with rigorous validation, we propose tiered reproducibility standards:

Bronze Standard (minimum for publication): List exact tickers, date ranges, data sources, and all filtering criteria. Report all hyperparameters tested and final selected values. Provide pseudocode for core algorithms. Disclose all package versions, programming language version, and random seeds.

Silver Standard (expected for computational papers): Bronze requirements plus a public repository with implementation code, a synthetic data generator reproducing key statistical properties of the dataset, and explicit documentation of the estimation window protocol.

Gold Standard (expected for flagship journals): Silver requirements plus complete replication scripts producing all tables and figures. Where proprietary data is used, provide anonymised or synthetic data with identical key statistical properties (mean, variance, autocorrelation structure, and, to the extent feasible, tail behaviour) enabling independent verification of the primary reported findings. We acknowledge that synthetic data cannot perfectly reproduce all statistical properties of proprietary datasets, particularly tail dependence and extreme-event regime structure. Additionally, Gold standard papers should provide containerised execution environments (e.g., Docker or Singularity images) specifying all software package versions, random seeds, and hardware environment.

E. The Future of Regime Detection: Paths Forward

For regime detection to mature into reliable methodology, several research directions are critical:

Formalising Conditions for Meaningful Regime Distinction: The most foundational obstacle to progress is not methodological but conceptual. As documented in Section III, the field has not yet established the conditions under which a claimed regime distinction carries statistical, temporal, and economic significance. Progress requires clarifying three conditions at minimum: the temporal scale at which a distinction must persist to qualify as a regime rather than transient noise, the degree of statistical separability required between regime-specific distributions, and the economic context in which the distinction is claimed to be actionable. Formalising these conditions is the precondition for all other progress.

Ground Truth Construction: Even without objectively true regimes, construct consensus labels through expert annotation, macroeconomic indicators, or ensemble methods (including through approaches building on [37], [38], [90]) to enable supervised validation and quantitative performance measurement.

Multi-Scale Integration: Develop frameworks coherently modelling regimes across time scales (intraday volatility states, multi-day trends, quarterly cycles) rather than treating each as independent [16], [17], [92].

Causal Validation: Move beyond correlational detection to causal analysis. Do detected regime changes predict future behaviour, or merely describe past data? The framework of [23] for regime-conditioned causal inference is a promising direction.

Portfolio Applications with Full Cost Accounting: Future work should apply regime frameworks to realistic portfolio problems with explicit transaction cost modelling, following the examples of [5], [57], [58], [62], [65], [66], [86], [89], [99], [104], [105].

Market-Specific Specialisation: Rather than forcing unified frameworks, develop specialised methodologies acknowledging cryptocurrency [47]–[51], equity [2], [15], [44], [45], FX [13], [52], [53], and macro regime detection as distinct subfields.

The path forward requires collective commitment to higher standards, transparent reporting, and genuine scientific rigour.

VIII. CONCLUSION

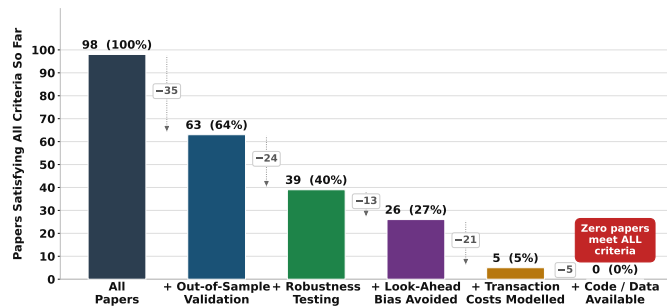


Fig. 9: The validation compliance funnel. Starting from all 98 papers, each bar shows how many papers satisfy all criteria applied so far. The cascade from 98 to 0 is the paper’s central empirical finding: no paper in the regime detection literature simultaneously satisfies out-of-sample validation, robustness testing, look-ahead bias avoidance, transaction cost modelling, and code/data availability.

This survey has examined 98 peer-reviewed papers spanning more than three decades of market regime detection research. The central empirical finding, captured in Figure 9, is stark: successive application of five minimum quality criteria reduces the compliant set to zero. The field has grown explosively, with 68 papers (69.4%) published since 2020, yet three structural failures have deepened rather than resolved. Ontologically, regime definitions are almost universally absent, with only one paper providing operational classification rules. Epistemologically, five paradigm families embody conflicting theoretical commitments that render their results formally incomparable even on identical data. Pragmatically, the disconnect between statistical performance and economic utility is pervasive, with transaction costs ignored in the vast majority of papers and post-cost advantages typically negligible when they are finally accounted for.

What does this mean in practice? *For a PhD student* choosing a dissertation topic, the implication is not that regime detection is exhausted but that the field’s foundational

problems remain genuinely open. The most valuable contributions will not be yet another regime classifier applied to a novel dataset; they will be work that answers the three questions this survey has identified as unresolved: what formalises the distinction between a regime and transient noise, under what conditions do detected regimes provide actionable economic value net of costs, and how should methods be selected when different paradigms are incommensurable by design. A dissertation that rigorously tackles any one of these would advance the field more than a dozen incremental modelling papers. The controlled case study in Section VI-C suggests regime-transition velocity as one such formalisation: a quantifiable, asset-specific property that predicts whether a detection method will generate economic value or destroy it, independent of classification accuracy.

For a journal editor setting review criteria, the survey’s validation funnel translates directly into a checklist. Accepting papers that lack out-of-sample evaluation, ignore transaction costs, or provide no code perpetuates exactly the reproducibility deficit that prevents cumulative knowledge development. The tiered reproducibility standards proposed in Section VII are operationalisable as desk-rejection criteria: Bronze standard as the minimum for any computational submission, Silver for papers making strategy claims, Gold for flagship journals. Requiring a meaningful baseline, at minimum buy-and-hold or a single-regime GARCH model evaluated on an identical out-of-sample period net of costs, would alone eliminate a substantial portion of the literature’s illusory performance gains.

For a practitioner deciding whether to trust published results, the appropriate prior should be sceptical. The literature systematically overstates regime detection value for predictable structural reasons: look-ahead contamination inflates backtested performance, absent transaction costs make strategies appear profitable that are not, and the absence of shared benchmarks means reported outperformance is almost always relative to other regime models rather than to simple alternatives the practitioner already has access to. The honest papers in this corpus, those that report results net of costs, with genuine out-of-sample evaluation, and compared against passive benchmarks, consistently find advantages that are modest at best and that vanish entirely at high transition frequencies. A practitioner applying a regime model to a high-velocity asset such as cryptocurrency should, on the basis of the evidence here, treat the results of any academic paper as an upper bound on achievable performance before accounting for execution realities.

The infrastructure required for progress exists. Shared benchmark datasets, the February–May 2020 period as a mandatory stress-test window, information-based sampling schemes as a robustness check, and asset-class-specific evaluation frameworks are all implementable without new theoretical machinery. What the field requires is the methodological discipline to apply standards that are already understood and the collective willingness to report results that may be less flattering than the current literature suggests.

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